

A drug from poison: how the therapeutic effect of arsenic trioxide on acute promyelocytic leukemia was discovered

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It is surprising that, while arsenic trioxide (ATO) is now considered as “the single most active agent in patients with acute promyelocytic leukemia (APL)”, the most important discoverer remains obscure and his original papers have not been cited by a single English paper. The discovery was made during the Cultural Revolution when most Chinese scientists and doctors struggled to survive. Beginning with recipes from a countryside practitioner that were vague in applicable diseases, Zhang TingDong and colleagues proposed in the 1970s that a single chemical in the recipe is most effective and that its target is APL. More than 20 years of work by Zhang and colleagues eliminated the confusions about whether and how ATO can be used effectively. Other researchers, first in China and then in the West, followed his lead. Retrospective analysis of data from his own group proved that APL was indeed the most sensitive target. Removal of a trace amount of mercury chloride from the recipe by another group in his hospital proved that only ATO was required. Publication of Western replication in 1998 made the therapy widely accepted, though neither Western, nor Chinese authors of English papers on ATO cited Zhang’s papers in the 1970s. This article focuses on the early papers of Zhang, but also suggests it worth further work to validate Chinese reports of ATO treatment of other cancers, and infers that some findings published in Chinese journals are of considerable value to patients and that doctors from other countries can benefit from the clinical experience of Chinese doctors with the largest population of patients.

leukemia, arsenic trioxide, acute promyelocytic leukemia

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Acute promyelocytic leukemia (APL) had been one of the most aggressive and fatal forms of acute leukemia, but is now one of the most treatable form of leukemia. While there are still room for improvement in treating APL, significant progress has been made in the last few decades by the applications of cytarabine (arabinosyl cytosine, Ara-C) [1], anthracyclines [2–4], arsenic trioxide (As₂O₃, ATO) [5–8] and all-trans retinoic acid (ATRA) [9,10].

The discoveries of cytarabine, anthracyclines and ATRA are well known, whereas the history of the discovery of

ATO therapy remains unknown. Most APL researchers cited papers published in the 1990s, which were 20 years later than the original papers. Some authors also seem to be confused about the original discoverer of ATO. These mistakes are regrettable when ATO is now considered to be “the most biologically active single drug in APL” by a panel of International Leukemia Experts for the European LeukemiaNet [11] or “the single most active agent in patients with APL” [12] and that the combination of ATO and ATRA holds the promise to “replace conventional approaches for most, if not all, patients in the very near future” [12]. The past decade has witnessed the general acceptance of ATO

以毒攻毒：三氧化二砷治疗急性早幼粒细胞白血病的疗效是如何被发现的

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令人惊讶的是, 尽管三氧化二砷 (ATO) 现在被认为是“治疗急性早幼粒细胞白血病 (APL) 患者最有效的单一药物”, 但最重要的发现者仍然默默无闻, 他的原始论文从未被任何一篇英文论文引用。这项发现是在文化大革命期间做出的, 当时大多数中国科学家和医生都在为生存而挣扎。从一位乡村医生的模糊适用疾病的药方开始, 张亭栋和同事在20世纪70年代提出, 药方中的单一化学物质最有效, 其靶点是APL。张和同事20多年的工作消除了关于ATO是否以及如何有效使用的困惑。其他研究人员, 首先在中国, 然后是西方, 都追随了他的脚步。对他自己团队的数据进行回顾性分析证明, APL确实是最敏感的靶点。他所在医院的另一个团队从药方中去除了微量的氯化汞, 证明只需要ATO。1998年西方复制的发表使该疗法被广泛接受, 尽管无论是西方还是英文论文的中国作者都没有引用张在20世纪70年代的论文。本文重点介绍张的早期论文, 但也建议值得进一步工作来验证中国关于ATO治疗其他癌症的报道, 并推断在中国期刊上发表的一些发现对患者具有相当大的价值, 并且其他国家的医生可以从中国医生拥有最大人口的临床经验中受益。

leukemia, arsenic trioxide, acute promyelocytic leukemia

Citation: 饶毅, 李瑞华, 张大庆. 毒物变良药: 三氧化二砷治疗急性早幼粒细胞白血病的发现过程. *中国科学: 生命科学*, 2013, 56: 495–502, doi: 10.1007/s11427-013-4487-z

急性早幼粒细胞白血病 (APL) 曾是最具侵袭性和致命性的急性白血病之一, 但现在已成为最易治疗的白血病之一。尽管 APL 的治疗仍有改进空间, 但在过去几十年中, 通过应用阿糖胞苷 (arabinosyl cytosine, Ara-C)[1], 蒽环类药物 [2–4], 三氧化二砷 (As₂O₃, ATO)[5–8] 和全反式维甲酸 (ATRA) [9,10], 取得了显著进展。

阿糖胞苷、蒽环类药物和 ATRA 的发现广为人知, 而

<style id='1'>从毒药到良药: 三氧化二砷治疗急性早幼粒细胞白血病的疗效是如何被发现的</style>ATO疗法仍然不为人知。大多数APL研究人员引用的论文发表于20世纪90年代, 比最初的论文晚了20年。一些作者似乎也对ATO的最初发现者感到困惑。当ATO现在被欧洲白血病网[11]的国际白血病专家小组认为是“APL中最具生物活性的单一药物”或“APL患者中最活跃的单一药物”[12], 并且ATO和ATRA的组合有望“在不久的将来取代大多数(如果不是所有)患者的传统方法”[12]时, 这些错误是令人遗憾的。过去十年见证了ATO的普遍接受

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[13] and accumulating proof of more ATO applications from relapsed APL to newly diagnosed APL [14,15].

Ignorance of, and confusions about, the early work result in part from the fact that the original papers were published in the Chinese language and in journals that are obscure even to most Chinese readers, although there are other factors of a more complex nature. This article summarizes the early papers from the discovery of ATO treatment of leukemia to the general acceptance of the use of ATO in treating APL. We also provide a list of the early papers in English for Western authors to use in citations.

We note that the authors of the present article have not worked on leukemia and write this article in our roles as researchers in the history of science. We hope that those working on leukemia can go into more details of the original work.

1 Cultural milieu and historical peculiarities of the discovery

In the 1960s and early 1970s, China was in a political turmoil known as the Great Proletarian Cultural Revolution. Although its origins lie in the political intentions of Mao ZeDong, the supreme leader of China at the time, the Cultural Revolution has affected more than one generation of Chinese people directly and indirectly. Some would view what is happening in China now in part as a consequence of, or a reaction to, the Cultural Revolution, with most viewing the Cultural Revolution as negative, if not disastrous.

Of the many leftist policies during the Cultural Revolution, most were harmful, but some had mixed or even positive effects intended or unintended by the policy maker(s). One policy directly related to Mao was to improve the medical conditions of rural China by sending doctors from urban hospitals to the countryside in “Circulating Medical Teams”. Doctors in such teams will go to multiple villages and members of the team will rotate. Another policy was to emphasize the importance of traditional Chinese medicine and drugs. The interception of these two led to many claims of findings of great effects of some Chinese medicine or treatment, most of which were abandoned within a few years. However, a few have withstood the test of time. The discovery of ATO was one such example.

Arsenic has been used for a long time, both in China and in the West. Several traditional Chinese medical recipes contain arsenic, but they were combinations of multiple chemicals with unclear targets. Western uses of arsenic were also ill-defined [16]. For example, Thomas Fowler of Britain invented a solution containing potassium arsenite (KAsO₂) in 1786, and used it for agues, remittent fevers and periodic headaches [16]. After leukemia was discovered in 1845, Fowler’s solution was used in treating leukemia in 1865, and again in 1931 [17]. Arsenic and irradiation were the main forms of treatment for chronic myelocytic leuke-

mia (CML) until 1953, when they were replaced by chemotherapy with busulfan [16,17]. Arsenic treatment of leukemia was no longer a standard drug for leukemia from then on. In China, Guan JiRen, a doctor in Harbin Medical University, tried to use Fowler’s solution to treat leukemia in 1958 and came to the conclusion that it was ineffective [18]. In the 1950s and 1960s, Zhou AiXiang in Beijing and Gu DeQing in Shanghai used combinations that included arsenic sulfate to treat leukemia [19]. It is also doubted that arsenic sulfate can be turned into ATO because the preparation of the Chinese medicine did not involve the high temperature required for the conversion of arsenic sulfate into ATO. In 1972, a report appeared in *Anti-Cancer Battle News of Liaoning Province*, a publication explicitly labeled as “internally circulated materials” under the authorship of Chaoyang People’s Hospital Department of Pediatrics [20]. It reported the treatment of 16 cases of acute granulocytic leukemia children by a combination of arsenic and chemotherapy. Because this treatment did not separate arsenic from the chemotherapy available then, it was unclear whether arsenic was helpful as an addition to chemotherapy, nor was it known how effective the combination was: success rate among the 16 patient was not reported and blood analysis was shown for only one patient [20]. A 1974 review by the Chinese Academy of Traditional Medicine summarized different approaches and medicines for leukemia treatment by traditional Chinese medicine listed As₂S₃ and toad venom (and others, including pure chemicals of Western invention) [21]. It cited [19] and [20] as supporting the use of As₂S₃. It should be noted that neither Gu nor Zhou has reduced their recipes to a single component, even today [22]. Because the preparation and processing of the Chinese medicine containing As₂S₃ did not involve the high temperature required for the conversion of arsenic sulfate into ATO, the relation of arsenic sulfate with ATO in such recipes is unclear. The 1974 review did not provide a conclusive recommendation on the type of Chinese medicine for the treatment of leukemia. Toad venom and quite a number of other medicines discussed in that review have not become a standard therapy for any type of leukemia. In summary, by 1974, it was unclear which of the traditional Chinese drugs can be used to treat leukemia and the effectiveness of those tested by then were uncertain.

In the early 1970s, Han TaiYun, a pharmacist of the First Affiliated Hospital of Harbin Medical University, was a member of a circulating medical team. He learned that a countryside practitioner of traditional Chinese medicine used a combination of arsenic, mercury and toad venom to treat lymphatic tuberculosis and cancers. In March 1971, Han made a solution that contained these three components, which he called “713” (after the year and month of his preparation) or “Ailin (literally meaning cancer effective)”. Intramuscular injection showed effects in some cancer patients. The “713” solution was hotly sought after locally for a while but faded soon from the public because of its tox-

[13] 以及从复发性APL到新诊断APL的更多ATO应用不断积累的证据 [14,15].

对早期工作的忽视和混淆, 部分原因在于原始论文是用中文发表的, 而且发表在即使对大多数中国读者来说也鲜为人知的期刊上, 尽管还有其他更复杂的因素。本文总结了从发现ATO治疗白血病到普遍接受使用ATO治疗APL的早期论文。我们还提供了一份早期英文论文清单, 供西方作者引用。

我们注意到, 本文作者并未从事白血病研究, 撰写本文是作为科学史研究者的角色。我们希望从事白血病研究的人员能够更详细地探讨原始工作。

1 发现的文化背景和历史特殊性

在20世纪60年代和70年代初, 中国正处于一场被称为无产阶级文化大革命的政治动荡之中。尽管其起源在于当时中国最高领导人毛泽东的政治意图, 但文化大革命直接或间接地影响了不止一代中国人。有些人会将中国现在发生的事情部分视为文化大革命的后果或反应, 大多数人认为文化大革命是负面的, 甚至可以说是灾难性的。

在文化大革命期间的众多左倾政策中, 大多数是有害的, 但有些政策产生了混合甚至积极的效果, 无论是政策制定者有意还是无意的。其中一项与毛泽东直接相关的政策是通过派遣城市医院的医生到农村组成“巡回医疗队”来改善中国农村的医疗条件。这些医疗队的医生会走访多个村庄, 队员也会轮换。另一项政策是强调中医药的重要性。这两项政策的交叉导致了許多关于某些中药或治疗方法具有巨大疗效的说法, 其中大部分在几年内就被放弃了。然而, 少数经受住了时间的考验。三氧化二砷 (ATO) 的发现就是这样一个例子。

砷在东方和西方都有着悠久的历史。一些中药方剂中含有砷, 但它们是多种化学物质的组合, 目标不明确。西方对砷的使用也定义不清 [16]。例如, 英国的托马斯·福勒于1786年发明了一种含有亚砷酸钾 (KAsO₂) 的溶液, 并将其用于治疗疟疾、弛张热和周期性头痛 [16]。1845年白血病被发现后, 福勒溶液于1865年再次被用于治疗白血病 1931 [17]。砷和放射治疗是治疗慢性粒细胞白血病的主要形式, 直到1953年, 它们被白消安化疗取代。

白血病 (CML) 的治疗, 直到1953年, 它们被白消安化疗取代 [16,17]。从那时起, 砷治疗白血病不再是白血病的标准药物。在中国, 哈尔滨医科大学的医生关继仁于1958年尝试使用福勒溶液治疗白血病, 并得出结论认为其无效 [18]。在20世纪50年代和60年代, 北京的周爱祥和上海的顾德清使用了包含硫化砷的组合来治疗白血病 [19]。人们也怀疑硫化砷能否转化为三氧化二砷 (ATO), 因为中药的制备不涉及将硫化砷转化为三氧化二砷所需的高温。1972年, 一份报告发表在《辽宁省抗癌战报》上, 这是一份明确标注为“内部流通材料”的出版物, 作者是朝阳市人民医院儿科 [20]。该报告报道了16例急性粒细胞白血病儿童通过砷和化疗联合治疗的情况。由于这种治疗没有将砷与当时可用的化疗分开, 因此不清楚砷作为化疗的补充是否有帮助, 也不知道这种组合的疗效如何: 16名患者的成功率没有报告, 血液分析只显示了一名患者 [20]。1974年, 中国中医研究院的一篇综述总结了中医治疗白血病的不同方法和药物, 列出了As₂S₃和蟾蜍毒 (以及其他, 包括西方发明的纯化学品) [21]。它引用了 [19] 和 [20] 作为支持使用As₂S₃的依据。值得注意的是, 即使是今天, 顾和周也没有将他们的方剂简化为单一成分 [22]。由于含有As₂S₃的中药的制备和加工不涉及将硫化砷转化为三氧化二砷所需的高温, 因此硫化砷与三氧化二砷在这些方剂中的关系尚不清楚。1974年的综述没有对治疗白血病的中药类型提供明确的建议。蟾蜍毒和该综述中讨论的许多其他药物尚未成为任何类型白血病的标准疗法。总而言之, 到1974年, 尚不清楚哪些中药可以用于治疗白血病, 并且当时已测试的那些药物的有效性也不确定。

20世纪70年代初, 哈尔滨医科大学第一附属医院药剂师韩太云是巡回医疗队的一员。他了解到一位乡村中医使用砷、汞和蟾蜍毒液的组合来治疗淋巴结核和癌症。1971年3月, 韩太云配制了一种含有这三种成分的溶液, 他称之为“713” (取自他配制溶液的年份和月份) 或“癌灵”。肌肉注射对一些癌症患者显示出疗效。“713”溶液一度在当地受到热捧, 但很快因其毒性而淡出公众视野。

icity. The target diseases of 713 were undefined, nor were the active chemical in 713.

Zhang TingDong was a doctor in the same hospital as Han. Zhang was born in 1932 and graduated in the 1950s from Harbin Medical University, after studying the regular (Western) medicine. He took classes of traditional Chinese medicine in the 1960s. He worked in the Department of Traditional Chinese Medicine of the First Affiliated Hospital of Harbin Medical University. He was initially asked by the Health Bureau of the Heilongjiang province to examine the validity of the claims of the countryside practitioner and later collaborated with Han.

2 Original discovery of the effective treatment of APL by ATO: 1973–1979

After 1972, Zhang and colleagues focused their research on leukemia. They also analyzed the components of “713” and suggested that arsenic was solely responsible for the therapeutic effect, whereas mercury caused kidney toxicity and toad venom caused hypertension. Neither of the latter two was therapeutically useful for leukemia. From then on, their recipe of Ailin I was mainly arsenic trioxide and a trace amount of mercury (at a ratio of 100:1 by weight), without toad venom.

The first paper by Zhang and Han was published in 1973 in a local Chinese journal. Zhang TingDong, Zhang PengFei, Wang ShouRen and Han TaiYun reported that they had used “Ailin solution” (also known as “Ailin I”) to treat six cases of chronic granulocytic leukemia [5]. They explicitly stated that the components of the solution were ATO and a trace amount of mercury chloride. All six patients improved after the treatment. They also mentioned that acute leukemia patients were being treated, but with no results in that paper.

In 1974, under the collective institutional authorship of the Department of Traditional Chinese Medicine and the Department of Laboratory Medicine of Harbin Medical University, they published a report in the university journal [6], summarizing the treatment of 17 cases of leukemia patients from January 1973 to April 1974. After going through different types of leukemia, they reported that Ailin I was effective in treating multiple types of leukemia, leading to complete remission (CR) in acute leukemia patients. In 1976, they used an institutional authorship to publish a report on five cases of acute leukemia in which they had achieved CR.

In 1979, Rong FuXiang and Zhang published two cases of acute granulocytic leukemia, one with CR for four and a half years and the other for three years [7].

A second paper by Zhang and Rong in 1979 summarized their treatment results from 55 cases of acute leukemia [8]. Twenty-three leukemia patients were treated with Ailin I alone (from 1973 to 1974), 20 were treated with Ailin I in

combination with Western chemotherapy and other Chinese medicines from 1975 to 1976, and 12 treated with Ailin I plus other Chinese medicines and chemotherapy from 1977 to 1978. For each patient, they presented leukemia subtypes and clinical observations. All 55 cases improved to some extent, with a remission rate of 70% and with CR in 12 cases. Side effects were small with the doses they used. They then applied 10 times the equivalent of what they used for adult human patients to 12 rabbits. No toxicity was observed in the heart, the liver, the spleen or the kidney of the rabbits.

While the 1973 paper reported their pioneering findings, the second 1979 paper represented their understanding of the therapeutic effect [8]. There are three important questions about early work of Zhang and colleagues: (i) Had they shown that the therapeutic effect came from Ailin I, but not from other Western chemicals or Chinese medicines? (ii) Had they realized that the effect of Ailin I came from ATO but not from mercury in the solution? (iii) Had they known the effect of ATO on APL?

Answers for all three questions can be found in [8] which explicitly stated that (i) significant improvement was observed in three patients (one adult and two children) using only Ailin I, but no other Western or Chinese drugs. At the time of publication, the children had survived for more than four years and the adult more than nine months. When using other Chinese medicines, Zhang and Rong pointed out that those were not used for treatment of leukemia, but for supporting the general health of the patients so that they could tolerate more treatments; (ii) the effective component of Ailin I was ATO (on page 11 of their paper); (iii) acute granulocytic leukemia (M3 type of the French-American British FAB classification, also known as APL) was the most sensitive to the treatment, which was a conclusion reiterated on pages 10 and 11 of their paper.

We can see that, by 1979, Zhang and his collaborators had clearly reached our current understanding that ATO could treat leukemia, especially that of the M3 subtype or APL.

3 Further studies by the Zhang group from the 1980s to the 1990s

In 1981, a paper under an institutional authorship with a footnote indicating Zhang as the supervisor (with eight other authors) reported 73 cases of acute granulocytic leukemia patients, with a CR of 24% and remission rate of 86% after Ailin I treatment [23]. In 1982, Zhang and Li presented a report to a national meeting on 22 cases of CR by Ailin I and on 98 cases of non-lymphatic leukemia. In 1982 and 1983, Zhang published reviews of his work [24,25].

In 1984, Zhang and Li [26] published a summary of 81 cases which they had treated since 1971. Among the 22 cases of CR, they pointed out that seven were of the M2

toxicity. 713's target diseases were undefined, nor were the active chemical in 713.

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2 Triarsine effectively treated APL: the first discovery: 1973-1979

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We can see that, by 1979, Zhang and his collaborators had clearly reached our current understanding that ATO could treat leukemia, especially that of the M3 subtype or APL.

3 Zhang TingDong's team's further research in the 1980s and 1990s

In 1981, a paper under an institutional authorship with a footnote indicating Zhang as the supervisor (with eight other authors) reported 73 cases of acute granulocytic leukemia patients, with a CR of 24% and remission rate of 86% [23]. In 1982, Zhang and Li presented a report to a national meeting on 22 cases of CR by Ailin I and on 98 cases of non-lymphatic leukemia. In 1982 and 1983, Zhang published reviews of his work [24,25].

In 1984, Zhang and Li [26] published a summary of 81 cases which they had treated since 1971. Among the 22 cases of CR, they pointed out that seven were of the M2

type and 15 were of the M3 type. They again stated that the effect on M3 type was particularly obvious. Zhang [27] published another paper on the effect of Ailin I on non-lymphatic acute leukemia.

In 1991, Sun et al. [28] continued the work of Zhang and Li [26]. They reported that Ailin I had been used to treat 32 APL cases from 1974 to 1985, with CR in 19 cases and that 16 cases had survived for more than five years. This confirms the high success rate for ATO treatment of APL.

In 1992, Sun et al. [29] published a short “Sharing Experience” paper, reviewing materials identical to the 1991 paper. Oddly, most English papers cite this 1992 paper for the discovery of ATO treatment of APL, although both papers were in Chinese.

Because Zhang TingDong’s papers from the 1970s to the early 1990s included a trace amount of mercury chloride, in addition to ATO (at a ratio of 1:100 by weight), strictly speaking, they had not proven that mercury chloride did not have a positive effect, despite the fact that their 1973 paper had mentioned that only ATO was the effective ingredient in Ailin I.

In 1995 and 1996, Zhang Peng and colleagues from the same hospital as Zhang TingDong published two papers which showed the effectiveness of ATO alone without mercury [30,31]. The 1995 paper was an abstract, which did not explicitly state that mercury chloride was not included in the 713 solution, although Zhang later said that they used only ATO. The 1996 paper did show that ATO, but not mercury chloride, was used. They treated 130 APL patients from 1992 to 1995, among which 72 went through one or more courses of treatment. A CR of 73% was observed in patients undergoing initial treatments and 52% in recurrent patients [31].

In the process of uncovering the history of ATO research, we found no evidence that leukemia classification by traditional Chinese medical theories was useful for discovering the target of ATO. Here we separate traditional Chinese medicine into drugs and theories. Had the traditional Chinese medical theories (CMTs) been helpful for developing ATO as a treatment for leukemia? Zhang and colleagues discussed five types of leukemia based on CMT classification, there was no difference of ATO on different CMT types [6–8]. In this regard, the Western classification of leukemia was helpful. When they completely gave up the CMT classification, the effect was more obvious. Interestingly, their first paper in 1973 did not mention CMTs, but their later papers in the 1970s and 1980s did. Lack of evidence for the utility of CMTs does not disprove the CMTs, but it is so far unclear whether the CMTs are important or essential for scientific studies of traditional Chinese drugs.

4 Chinese contributions to APL treatment

Anthracyclines (including daunorubin) and cytarabine be-

came frontline treatment for APL because of research in the West [1–3]. Chinese contributions in the discoveries of ATO and ATRA came after those, but have significantly improved APL treatment. Here we place the Chinese discoveries in the historical context.

In 1973, Zhang and colleagues of China reported the therapeutic effect of ATO on leukemia [5], and Zhang and Rong [8] in 1979 suggested that APL was particularly sensitive to ATO.

In 1977, Collins et al. [32] at the NCI successfully established a cell line (HL-60) from an APL patient. Collins et al. used it to screen for chemicals which could induce the differentiation of HL-60 cells to mature into normal cells. In 1980, Breitman et al. [33] discovered that all-trans retinoic acid (ATRA) and 13-cis retinoic acid induced HL-60 differentiation into mature cells. Related chemicals such as Vitamin A was 1000 fold less effective. They suggested that “this compound could provide a new therapeutic tool in the treatment of acute myeloid leukemia”.

In 1981, Breitman et al. [34] tested drug sensitivity of leukocytes from the peripheral blood of leukemia patients and found that cells induced to differentiate by ATRA all came from two patients with APL. Olsson and Brietman [35] in 1982 showed that retinoic acid could also induce U-937 lymphoma cells to differentiate. In 1983, Honma et al. [36] from Japan reported the effects of multiple chemicals on inducing differentiation of cells from different leukemia patients, and found that ATRA was among those capable of inducing the differentiation of leukocytes from APL patients. Koeffler [37] in 1983 summarized *in vitro* studies including cellular differentiation by retinoic acid and other chemicals, viewing ATRA and 13-cis retinoic acid as equivalent in differentiating APL leukocytes.

Single cases of APL treatment by 13-cis retinoic acid were reported by four groups: Flynn et al. [38] in 1983 from Minnesota, USA; Nilsson [39] in 1984 from Lund, Sweden; Daenen et al. [40] in 1986 from the Netherland; and Fontana et al. [41] in 1986 from west Virginia, USA.

In 1985, Wang Zhen-Yi of Shanghai Second Medical College could obtain ATRA (but not 13-cis retinoic acid) from a local source. He used it to successfully treat a five-year-old girl. In 1987, his group published a paper in the English edition of the *Chinese Medical Journal*, reporting the use of ATRA (alone or in combination) for the treatment of six APL patients [9]. In 1988, Wang’s group published their use of ATRA in the treatment of 24 APL patients in *Blood* [10]. It cited Breitman et al. [33,34] and Koeffler [37], which reported the effects of ATRA and 13-cis retinoic acid on inducing the differentiation of leukemia leukocytes, as well as Flynn et al. [38], Nilsson [39], Daenen et al. [40], and Fontana et al. [41] which reported treatment of APL patients by 13-cis retinoic acid.

Huang et al. [10], but not Huang et al. [9] (both in English, but the 1988 paper published in the US and the 1987 paper in China), drew international attention. Direct com-

型, 其中15例为M3型。他们再次指出, 对M3型的效果特别明显。张 [27]发表了另一篇关于艾林I对非淋巴性急性白血影响的论文。

1991年, 孙等 [28] 继续了张和李 [26]的工作。他们报告说, 从1974年到1985年, 艾林I已被用于治疗32例 APL病例, 其中19例获得完全缓解, 16例存活超过五年。这证实了ATO治疗APL的高成功率。

1992年, 孙等 [29]发表了一篇简短的“经验分享”论文, 回顾了与1991年论文相同的材料。奇怪的是, 大多数英文论文都引用了这篇1992年的论文来论证ATO治疗 APL的发现, 尽管这两篇论文都是中文的。

由于张亭栋在20世纪70年代到90年代初的论文中, 除了三氧化二砷 (ATO) 之外, 还含有微量的氯化汞 (按重量比1:100), 严格来说, 他们并未证明氯化汞没有积极作用, 尽管他们1973年的论文曾提到三氧化二砷是癌灵一号中唯一的有效成分。

1995年和1996年, 张亭栋同医院的张鹏及其同事发表了两篇论文, 表明单独使用三氧化二砷 (ATO) 而没有汞也有效 [30,31]。1995年的论文是一篇摘要, 没有明确说明713溶液中不含氯化汞, 尽管张鹏后来表示他们只使用了三氧化二砷。1996年的论文确实表明使用了三氧化二砷, 但没有使用氯化汞。他们治疗了1992年至1995年间的130名急性早幼粒细胞白血病 (APL) 患者, 其中72名患者接受了一个或多个疗程的治疗。在初治患者中观察到73%的完全缓解率 (CR), 在复发患者中观察到52%的完全缓解率 [31]。

在揭示三氧化二砷研究历史的过程中, 我们没有发现任何证据表明传统中医理论对白血病的分类有助于发现三氧化二砷的靶点。在此, 我们将传统中医分为药物和理论。传统中医理论 (CMT) 是否有助于将三氧化二砷开发为白血病治疗药物? 张氏及其同事讨论了基于中医分类的五种白血病类型, 三氧化二砷对不同中医类型 [6–8]没有区别。在这方面, 西医对白血病的分类是有帮助的。当他们完全放弃中医分类时, 效果更加明显。有趣的是, 他们1973年的第一篇论文没有提及中医理论, 但他们在1970年代和1980年代的后续论文中提及了。缺乏中医理论效用的证据并不能反驳中医理论, 但目前尚不清楚中医理论对于中药的科学研究是否重要或必不可少。

中国对APL治疗的4项贡献

蒽环类药物 (包括柔红霉素) 和阿糖胞苷

由于西方国家的研究, 成为APL的一线治疗药物 [1–3]。中国在ATO和ATRA发现方面的贡献晚于这些, 但显著改善了APL的治疗。在此, 我们将中国的发现置于历史背景中。

1973年, 中国张氏及同事报道了ATO对白血病的治疗作用 [5],, 张氏和荣氏于1979年提出APL对ATO特别敏感 [8]。

1977年, Collins等 [32] 在美国国家癌症研究所成功地从一名APL患者身上建立了一个细胞系 (HL-60)。Collins等用它来筛选能够诱导HL-60细胞分化成熟为正常细胞的化学物质。1980年, Breitman等 [33] 发现全反式维甲酸 (ATRA) 和13-顺式维甲酸能够诱导 HL-60分化为成熟细胞。相关化学物质如维生素A的效力低1000倍。他们提出“这种化合物可以为急性髓系白血病的治疗提供一种新的治疗工具”。

1981年, Breitman 等人 [34] 测试了白血病患者外周血白细胞的药物敏感性, 发现由 ATRA 诱导分化的细胞均来自两名 APL 患者。Olsson 和 Brietman [35]于 1982 年表明, 维甲酸也能诱导 U-937 淋巴瘤细胞分化。1983 年, 来自日本的 Honma 等人 [36]报告了多种化学物质对诱导不同白血病患者细胞分化的影响, 发现 ATRA 能够诱导 APL 患者白细胞分化。Koeffler [37] 于 1983 年总结了包括维甲酸和其他化学物质诱导细胞分化在内的体外研究, 认为 ATRA 和 13-顺式维甲酸在分化 APL 白细胞方面是等效的。

有四个研究组报告了 13-顺式维甲酸治疗 APL 的单例病例: 1983 年来自美国明尼苏达州的 Flynn 等人 [38]; 1984 年来自瑞典隆德的 Nilsson [39]; 1986 年来自荷兰的 Daenen 等人 [40]; 以及 1986 年来自美国西弗吉尼亚州的 Fontana 等人 [41]。

1985年, 上海第二医学院的王振义教授从当地获得了 ATRA (但不是13-顺式维甲酸)。他用它成功治疗了一名五岁女孩。1987年, 他的团队在英文版《中华医学杂志》上发表了一篇论文, 报道了 ATRA (单独或联合) 治疗六名 APL 患者的案例 [9]。1988年, 王教授的团队在《*Blood*》杂志上发表了他们使用 ATRA 治疗 24 名 APL 患者的报告 [10]。该报告引用了 Breitman 等人 [33,34] 和 Koeffler [37], 关于 ATRA 和 13-顺式维甲酸诱导白血病白细胞分化作用的报道, 以及 Flynn 等人 [38]、Nilsson [39]、Daenen 等人 [40], 和 Fontana 等人 [41] 关于 13-顺式维甲酸治疗 APL 患者的报道。

黄等人 [10], 而非黄等人 [9] (两篇均为英文, 但 1988 年的论文在美国发表, 1987 年的论文在中国发表), 引起了国际关注。直接的比较——

munication with French doctors also helped. The finding of Wang group with ATRA were soon replicated. In 1989, Chomienne et al. [42] from France compared the effects of ATRA and 13-cis retinoic acid with two APL patients for each chemical and felt that ATRA were more effective. In 1990, the same French group, after working with leukocytes from 22 APL patients *in vitro*, concluded that ATRA was 10 times more effective than 13-cis retinoic acid [43]. The effect of ATRA was also confirmed by Chinese doctors (e.g., Chen et al.) [44]. In 1991, Warrell et al. [45] in the US replicated the findings of the Wang group in China and the Degos group in France with nine out of 11 APL patients successfully treated by ATRA. Since then, ATRA was well recognized for APL treatment. In 1997, Tallman et al. [46] reported their studies of 346 APL patients, in which they compared the therapeutic effects of ATRA and the previously standard chemotherapy with daunorubin and cytarabine, and found it to be more effective if ATRA was used for both induction and maintenance.

In 1992, Duan et al. [47] published *in vitro* studies of the effect of ATO on leukemic cells. In 1995, Huang and coworkers from Dalian, China reported that a tablet with multiple components derived from traditional Chinese medicine (herbs and minerals) led to 98% CR in 60 APL patients [48]. One of the components contained arsenic disulfide.

In 1995 and February 1996, Zhang Peng and colleagues from Harbin reported their success in using ATO in achieving 73% of CR in 130 APL cases from 1992 to 1995. No cross-resistance was observed between ATO and ATRA [30,31].

In August 1996, Chen GuoQiang and 18 other authors (including Zhang TingDong in the middle and Chen Sai-Juan, Wang Zhen-Yi and Chen Zhu as the last authors) reported work from Shanghai Hematology Institute that used *in vitro* culture leukemic cells for mechanistic studies of the therapeutic effect of ATO on leukemia at the molecular level [49].

In 1997, Xu JingShu, Duan JingMian, Xu Ying, Xin XiaoMin, Song XiaoHong and Zhang TingDong reported a case of who had recurrent APL three times [50]. The patient was treated with Ailin I every time and had survived for 20 years.

In 1997, Chen et al. [51] from Shanghai published dose-dependent effect of ATO on leukemic cells *in vitro*. Shen et al. [52] from Shanghai reported that they used pure ATO to treat 15 APL patients, among which 10 cases with only ATO. CR was achieved in 90%.

In 1998, Soignet et al. [53] from the Memorial Sloan-Kettering Cancer Hospital and Cornell Medical College reported in the *New England Journal of Medicine* that they had treated 12 recurrent APL cases with ATO and observed CR in 11 cases. The mechanisms were thought to be partial cellular differentiation and apoptosis.

The Soignet et al. paper helped general international acceptance of ATO as a treatment of APL, which could not be

achieved by many papers published in China by Chinese doctors over the previous two decades.

5 Lack of recognition

ATO has now been well accepted (and generally used) nationally and internationally, saving lives in China and other countries. However, the discoverer remains largely unknown in academic and medical communities, although there was a 2001 story about him in the *New York Times* [54]. It is more striking that, while the discovery of the effect of ATRA on APL has led to both national and international awards to Wang Zhen-Yi, not a single national or international award has been given to Zhang TingDong or any of his Harbin colleagues for the discovery of the therapeutic effect of ATO on APL, although ATO was discovered more than a decade before ATRA and is recognized as “the most biologically active single drug in APL” by International Leukemia Experts for the European LeukemiaNet [11].

The reason for lack of recognition is not due to controversies. There was a patent contention by Hongde Sun, a member of Zhang’s group. It was quite late and the judge ruled in Zhang’s favor. Peng Zhang insisted that he was the first to show the effect of ATO alone, without mercury. Zhang and Rong [8] had suggested that ATO alone was effective, but they had not shown data with ATO alone. Both Sun and Zhang Peng have made important contributions, but it is clear that Zhang TingDong has played an undisputed key role in his persistent work from the 1970s to the early 1990s which turned the often fuzzy arsenic treatment with variable results into practically useful treatment with beneficial effects.

In 1998, Chen et al. [55] published in a Chinese journal that “from the early 1970s, Harbin Medical University discovered through clinical practices that arsenic trioxide could effectively treat APL. In the past two years, we have collaborated with HMU and used arsenic trioxide solution to treat APL patients resistant to ATRA and conventional chemotherapy”, affirming the work and priority of Harbin, though Zhang’s name and papers in the 1970s did not appear in the review [55].

Almost no English paper realized that Zhang had published his findings from 1973 to 1979. English papers, including those by Chinese scholars, only cited Sun et al. [29] and sometimes Zhang Peng et al. [31] as the first paper(s) for ATO treatment of leukemia. For example, Soignet et al. [53], which replicated the findings of Zhang in the 1970s and played a major role in the international acceptance of ATO treatment for APL, mentioned “recent Chinese reports” of CR in APL by ATO, and cited only Sun et al. [29], Zhang Peng et al. [31], and Shen et al. [52]. It is impossible to know from the Soignet et al. paper that the original findings were made in the 1970s by Zhang TingDong because both the tone and the citations made it seem that Chinese

<style id='1'></style>在过去的二十年里, 中国医生在中国发表的许多论文都取得了这一成就。

5 缺乏认可

ATO 现已在国内和国际上被广泛接受 (并普遍使用), 在中国和其他国家挽救了生命。然而, 尽管《纽约时报》[54] 在 2001 年曾报道过他, 但这位发现者在学术界和医学界仍然鲜为人知。更令人震惊的是, 虽然 ATRA 对 APL 作用的发现使王振义获得了国内和国际奖项, 但张亭栋或他的哈尔滨同事们却从未因发现 ATO 对 APL 的治疗作用而获得任何国内或国际奖项, 尽管 ATO 的发现比 ATRA 早十多年, 并且被欧洲白血病网络国际白血病专家认定为“APL 中生物活性最强的单一药物” [11]。

未获认可的原因并非争议。张氏团队成员孙宏德曾提出专利争议。该争议发生得相当晚, 法官裁定张胜诉。张鹏坚持认为他是第一个单独展示三氧化二砷 (ATO) 效果的人, 且不含汞。张和荣 [8] 曾提出单独使用三氧化二砷有效, 但他们没有展示单独使用三氧化二砷的数据。孙和张鹏都做出了重要贡献, 但很明显, 张廷栋在 1970 年代到 1990 年代初期的持续工作中发挥了无可争议的关键作用, 他将结果不一、模糊不清的砷治疗转变为具有有益效果的实用治疗。

1998 年, 陈等人 [55] 在中国期刊上发表文章称, “自 1970 年代初, 哈尔滨医科大学通过临床实践发现三氧化二砷可有效治疗 APL。在过去两年中, 我们与哈医大合作, 使用三氧化二砷溶液治疗对 ATRA 和常规化疗耐药的 APL 患者”, 肯定了哈尔滨的工作和优先权, 尽管张的名字和 1970 年代的论文并未出现在该综述中 [55]。

几乎没有英文论文意识到张亭栋在 1973 年至 1979 年间发表了发现。英文论文, 包括那些由中国学者撰写的论文, 只引用了孙等人 [29], 有时还引用了张鹏等人 [31] 作为 ATO 治疗白血病的第一篇论文。例如, Soignet 等人 [53], 复制了张亭栋在 1970 年代的发现, 并在 ATO 治疗 APL 的国际接受度中发挥了重要作用, 他们提到了 ATO 治疗 APL 的“近期中国报告”中的 CR, 并且只引用了孙等人 [29]、张鹏等人 [31], 和沈等人 [52]。从 Soignet 等人的论文中不可能知道最初的发现是张亭栋在 1970 年代做出的, 因为无论是语气还是引文都让人觉得中国

与法国医生的交流也起到了帮助作用。王振义团队关于 ATRA 的发现很快得到了复制。1989 年, 法国的 Chomienne 等人 [42] 比较了 ATRA 和 13-顺式维甲酸对两名 APL 患者 (每种化学物质各一名) 的疗效, 认为 ATRA 更有效。1990 年, 同一个法国团队在对 22 名 APL 患者的白细胞进行体外研究后, 得出结论: ATRA 的疗效是 13-顺式维甲酸的 10 倍 [43]。ATRA 的疗效也得到了中国医生 (如陈竺等人) 的证实 [44]。1991 年, 美国的 Warrell 等人 [45] 复制了中国王振义团队和法国 Degos 团队的发现, 用 ATRA 成功治疗了 11 名 APL 患者中的 9 名。自此, ATRA 在 APL 治疗中得到了广泛认可。1997 年, Tallman 等人 [46] 报告了他们对 346 名 APL 患者的研究, 其中他们比较了 ATRA 与之前标准的柔红霉素和阿糖胞苷化疗的治疗效果, 发现如果 ATRA 用于诱导和维持治疗, 效果会更好。

1992 年, Duan 等人 [47] 发表了 ATO 对白血病细胞体外作用的研究。1995 年, 来自中国大连的 Huang 及其同事报告称, 一种含有多种中药 (草药和矿物质) 成分的片剂使 60 名 APL 患者的 CR 率达到 98% [48]。其中一种成分含有二硫化砷。

1995 年和 1996 年 2 月, 来自哈尔滨的张鹏及其同事报告了他们使用 ATO 在 1992 年至 1995 年间使 130 例 APL 患者的 CR 率达到 73% 的成功经验。ATO 和 ATRA 之间未观察到交叉耐药性 [30,31]。

1996 年 8 月, 陈国强等 18 位作者 (包括中间的张亭栋和最后的陈赛娟、王振义、陈竺) 报告了上海血液学研究所的工作, 该工作使用体外培养的白血病细胞, 从分子水平上对 ATO 治疗白血病的机制进行了研究 [49]。

1997 年, 徐景书、段景绵、徐瑛、辛晓敏、宋晓红和张亭栋报道了一例 APL 复发三次的病例 [50]。该患者每次都用癌灵一号治疗, 并已存活 20 年。

1997 年, 上海的陈竺等人 [51] 发表了三氧化二砷对白血病细胞体外的剂量依赖性作用。上海的沈志祥等人 [52] 报道他们用纯三氧化二砷治疗了 15 例 APL 患者, 其中 10 例仅用三氧化二砷治疗。90% 的患者达到了完全缓解。

1998 年, 索瓦涅特等人 [53] 来自纪念斯隆-凯特琳癌症医院和康奈尔医学院, 在《新英格兰医学杂志》上报道他们用三氧化二砷治疗了 12 例复发性 APL 病例, 其中 11 例观察到完全缓解。其机制被认为是部分细胞分化和凋亡。

Soignet 等人的论文帮助 ATO 作为 APL 治疗方法获得了国际普遍认可, 而这在之前二十年间中国医生发表的许多论文中未能实现。

discoveries were made in the 1990s.

A 1996 news report in *Science* did mention Zhang [56], but stated that Zhang published his paper in 1992.

Zhang has published few English papers. In 2001, he and Chen GuoQiang were co-first authors (with Wang Zhen-Yi, Chen SaiJuan in the middle and Chen Zhu as the corresponding author) of a review about ATO in an international journal *Oncogene* [57]. In the introduction, they stated “recent” studies of ATO treatment of APL, citing the 1996 paper of Chen et al. [49]. On the second page, they stated that the research on ATO began in 1971, without citing any publications, and that they had treated more than a thousand patients of different types of cancers including “chronic granulocytic leukemia, lymphoma, esophageal cancer, and particularly APL”, but again without citing any literature. Thus, Zhang, presumably as the first author of an English paper, neglected to cite his own early papers, effectively burying the pioneering findings in the 1970s.

In 2002, Zhu et al. [58] published a review in *Nature Reviews Cancer*. In the figure illustrating milestones in APL treatments, Zhang TingDong in the 1970s were placed, but the citation in the text was Sun et al. [29] and the explanation in the reference list credited Sun et al. [29] as “first report of As₂O₃ therapy in APL”.

Both Sun et al. [29] and Zhang et al. [31] were published in Chinese, and neither of them cited papers from the 1970s. Thus, even if any international scholars attempted to obtain English translations of the 1992 and 1996 papers, they would not know the original 1970 papers.

In 2008, Wang Zhen-Yi and Chen Zhu [59] reviewed progress in APL treatment in *Blood*, with its first citation of ATO as the Zhu et al. [58] review, and further citations for ATO treatment of APL being Sun et al. (1992) [29], Zhang et al. (1996) [31], Chen et al. (1996) [49], Shen et al. (1997) [52] and Niu et al. (1999) [60].

In 2011, Chen et al. [61] published a review of the therapeutic effect of ATO on leukemia, which stated “in the early 1970s, a group from Harbin Medical University in northeastern China tested Ailing-1 containing 1% ATO and a trace amount of mercury chloride in a variety of cancers by intravenous administration”, without mentioning the researchers or citing papers of the 1970s. The review then cited Sun et al. (1992) [29] for showing that “Ailing-1 induced CR in 21 of 32 patients with APL with an impressive 10-year survival rate of 30%” before stating that “the efficacy of pure ATO in treating relapsed APL was then reported by Shanghai Institute of Hematology (SIH) in 1996–1999”, citing Shen et al. (1997) [52] and Niu et al. (1999) [60], ignoring the papers of Zhang Peng et al. from Harbin Medical University who had published a paper in 1995 and one in 1996 [30,31], which stated that they had used ATO alone (without a trace amount of mercury chloride).According to a 2013 blog by Zhang Peng, his results

were known to Chen and others who attended a national meeting in 1995 [62].

A 2011 paper on a 10-year follow-up study of ATO treatment of APL published by Au et al. [63] from Hong Kong cited a paper published by US authors in 2001.

Thus, no authors who can read Chinese have cited any of the 1970s papers in their English publications. In the current climate that English authors do not go to original papers in even French or German, it is no wonder that they do not know the titles and authors of the original papers published in Chinese. The contributions of Zhang TingDong and the precise timing of his discovery are therefore virtually unknown in the international academic and medical communities.

6 Implications of paying attention to work published in Chinese

In the past, there is a general negligence of clinical studies carried out in China. Language is only part of the reason. With the large number of patients in China, many doctors in China have more clinical experience than most doctors in Western countries. Some, even though a small fraction of, Chinese doctors may have insights into treatments that they have only published in Chinese journals.

In the case of ATO, for example, Chinese doctors including Zhang and others have reported ATO treatment of multiple cancers, from liver, stomach and colon cancers to lymphomas [64–66]. These are worth further test and validation.

An indirect inference is that rigorous studies of components of Chinese medicine may lead to more discoveries. For example, drugs used tentatively by Chinese hospitals or marketed aggressively by Chinese companies (without prior stringent tests), may prove to be more powerful and specific after rigorous studies, and become more internationally acceptable and will eventually help more patients and save more lives.

One of the authors (Rao Yi) thanks Li LianDa of the Chinese Academy of Traditional Medicine for bringing him to the attention of two papers [18,20] and the work of Zhou AiXiang and Gu DeXin.

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<style id='1'>发现于20世纪90年代。</style>

《科学》杂志1996年的一篇新闻报道确实提到了张 [56],, 但称张于1992年发表了他的论文。

张发表的英文论文很少。2001年, 他与陈国强共同担任一篇关于三氧化二砷的综述的共同第一作者(王振义、陈赛娟为中间作者, 陈竺为通讯作者), 该综述发表在国际期刊《癌基因》上。在引言中, 他们提到了三氧化二砷治疗急性早幼粒细胞白血病的“近期”研究, 并引用了陈等人1996年的论文。在第二页, 他们指出三氧化二砷的研究始于1971年, 但没有引用任何出版物, 并且他们已经治疗了包括“慢性粒细胞白血病、淋巴瘤、食道癌, 特别是急性早幼粒细胞白血病”在内的上千名不同类型癌症患者, 但同样没有引用任何文献。因此, 张作为一篇英文论文的第一作者, 却忽略了引用他自己的早期论文, 有效地埋葬了20世纪70年代的开创性发现。

2002年, Zhu 等人 [58] 在《*NatureReviews Cancer*》上发表了一篇综述。在阐述APL治疗里程碑的图中, 提到了20世纪70年代的张廷栋, 但文中引用的却是Sun 等人 [29], 参考文献列表中的解释也将Sun 等人 [29] 归功于“首次报道As₂O₃ 治疗APL”。

Sun 等人 [29] 和 Zhang 等人 [31] 的论文都是用中文发表的, 而且都没有引用20世纪70年代的论文。因此, 即使有国际学者试图获取1992年和1996年论文的英文译本, 他们也无法得知最初的1970年论文。

2008年, 王振义和陈竺 [59] 在《*Blood*》杂志上回顾了APL治疗的进展, 首次引用ATO的文献是Zhu 等人 [58] 的综述, 随后引用ATO治疗APL的文献包括Sun 等人 (1992) [29]、Zhang 等人 (1996) [31]、Chen 等人 (1996) [49]、Shen 等人 (1997) [52] 和 Niu 等人 (1999) [60]。

2011년 Chen 등 [61] 은 ATO의 백혈병 치료 효과에 대한 리뷰를 발표했는데, 여기에는 “1970년대 초, 중국 북동부 하얼빈 의과대학의 한 그룹이 1% ATO와 미량의 염화수은을 함유한 Ailing-1 을 다양한 암에 정맥 투여 방식으로 테스트했다” 고 명시되어 있었지만, 연구자나 1970년대 논문에 대한 언급은 없었다. 이 리뷰는 이어서 Sun 등(1992) [29]의 연구를 인용하여 “Ailing-1이 APL 환자 32명 중 21명에게서 CR을 유도했으며, 10 년 생존율이 30% 에 달하는 인상적인 결과를 보였다” 고 언급한 후, “순수 ATO가 재발성 APL 치료에 효과가 있다는 것이 1996-1999년 상하이 혈액학 연구소(SIH)에 의해 보고되었다” 고 명시하며 Shen 등(1997) [52] 과 Niu 등(1999) [60], 의 논문을 인용했다. 이 과정에서 하얼빈 의과대학의 Zhang Peng 등이 1995년과 1996 [30,31], 에 발표한 논문(이 논문들에서는 ATO 단독 사용(미량의 염화수은 없이)을 명시하고 있었다)은 무시되었다. Zhang Peng의 2013년 블로그에 따르면, 그의 결과는

为参加 1995 [62]全国会议的陈等人所知晓。

Au 等人于 [63] 在香港发表的一篇关于ATO治疗APL的10年随访研究的2011年论文, 引用了美国作者于2001年发表的一篇文章。

因此, 没有能阅读中文的作者在其英文出版物中引用过20世纪70年代的任何论文。在当前英文作者甚至不查阅法文或德文原始论文的环境下, 他们不知道中文原始论文的标题和作者也就不足为奇了。因此, 张亭栋的贡献及其发现的确切时间在国际学术界和医学界几乎不为人知。

6 关注中文出版物的影响

过去, 人们普遍忽视在中国进行的临床研究。语言只是部分原因。由于中国患者数量众多, 许多中国医生比大多数西方国家的医生拥有更多的临床经验。一些(尽管只是一小部分)中国医生可能对治疗有独到的见解, 而这些见解只发表在中文期刊上。

以三氧化二砷为例, 包括张亭栋在内的中国医生已经报道了三氧化二砷治疗多种癌症的案例, 从肝癌、胃癌、结肠癌到淋巴瘤 [64–66]。这些都值得进一步的测试和验证。

一个间接的推论是, 对中药成分进行严谨的研究可能会带来更多的发现。例如, 中国医院试用或中国公司积极推销(未经严格前期测试)的药物, 在经过严谨研究后, 可能会被证明更有效、更具特异性, 从而更被国际接受, 并最终帮助更多患者, 挽救更多生命。

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1) Files of the major Chinese papers cited in this article can be accessed at the blog of the author: <http://blog.sciencenet.cn/blog-2237-658474.html>

1) 本文引用的主要中文论文文件可在作者博客查阅: <http://blog.sciencenet.cn/blog-2237-658474.html>

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